

Federated Prototype Learning with Privacy-Aware Multi-Metric Client Selection

for Software Defect Prediction

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Presentation Outline

- 1 Problem & Motivation
- 2 Background & Related Work
- 3 Proposed Solution: FPLPA + MultiMetric
- 4 Experiments & Results
- 5 Conclusions & Future Work

15-Minute Roadmap

0–2 min	Problem & Motivation
2–4 min	Background & Gaps
4–8 min	Proposed Solution
8–12 min	Experiments & Results
12–15 min	Conclusions & Q&A

Problem Statement

Why does this matter?

Software bugs cost the industry **\$2.4 trillion** annually. Early defect prediction is critical — but it is hard:

- Defect data is **private** (IP, regulations like GDPR)
- Different teams have **different metrics and distributions**
- Single-project data is **small and imbalanced**

Key Challenges

- **Privacy**: Raw data cannot leave each organization
- **Heterogeneity**: Different metric spaces and defect rates
- **Imbalance**: Only 5–20% modules are defective
- **Client Selection**: Which clients to include each round?

Core Research Question

How to collaboratively predict defects across organizations **without sharing raw data**, despite

Defect Rates in Our Data

CM1: 9.8% · MW1: 7.5%
PC1: 6.9% · Apache: 15.5%
FFEM: 11.5%

Federated Learning — A Quick Primer

Standard Federated Averaging (FedAvg)

Each round t :

- 1 Server selects subset S_t of clients
- 2 Clients receive global model w_t
- 3 Each client trains locally for E epochs
- 4 Clients send back updated weights
- 5 Server aggregates:

$$w_{t+1} = \frac{\sum_{k \in S_t} n_k w_{t+1}^{(k)}}{\sum_{k \in S_t} n_k}$$

Problem: Assumes IID data — violated in practice.

Existing Client Selection Methods

Random	Uniform random (baseline)
FedCS	Greedy top- k by data volume
PowerChoice	Top- k by highest local loss
MultiMetric	Our method: 4 signals

Research Gap

No prior method combines **learning progress + diversity + consistency + exploration** in a unified framework, especially for defect prediction with its unique imbalance challenges.

Why Prototypes? Privacy + Communication

Prototype = Mean Class Embedding

For client k , class c :

$$p_c^{(k)} = \frac{1}{|D_c^{(k)}|} \sum_{(x,y) \in D_c^{(k)}} f_\theta(x)$$

A 32-dimensional vector summarising all samples of that class.

Server Aggregation:

$$p_c^{\text{global}} = \frac{1}{|S_t|} \sum_{k \in S_t} p_c^{(k)}$$

Why Prototypes Beat Gradient Sharing

- **Privacy:** Coarser summary; harder to invert
- **Communication:** 32D vector vs. full model weights
- **Heterogeneity:** Model architectures can differ
- **Bound:** $I(p_c; x) \leq \frac{1}{|D_c|} H(x)$
— leakage shrinks as class size grows

Local Training Loss

$$\mathcal{L}_c = \mathbb{E}_{(x,y) \in D_c} \left[\frac{1}{2} \|f_\theta(x) - p_c^{\text{global}}\|^2 \right]$$

System Overview: FPLPA Framework

Four Core Components

1 MultiMetric Selector

Decides which k of N clients participate each round

2 Local CPN Training

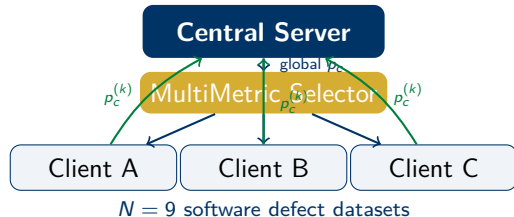
Client trains lightweight CNN with CE + prototype loss

3 Prototype Generation

Each client computes mean class embeddings

4 Server Aggregation

Server averages prototypes; sends back for next round



MultiMetric: The Core Contribution

Composite Score for Client i at Round t :

$$S_i^{(t)} = \underbrace{0.70 \bar{\Delta}_i}_{\text{progress}} + \underbrace{0.10 \text{cons}_i}_{\text{stability}} + \underbrace{0.20 \text{div}_i}_{\text{diversity}} + \underbrace{\beta \text{UCB}_i}_{\text{explore}}$$

Signal Definitions

$\bar{\Delta}_i$	EMA of loss drop ($\ell^{\text{before}} - \ell^{\text{after}}$)
cons_i	$\frac{1}{1+2\sigma_i}$ (stability of Δ)
div_i	0.3 if recently selected, else 1.0
UCB_i	$0.3 \sqrt{\frac{4 \ln t}{\text{count}_i + 1}}$

Annealed Softmax Selection:

$$P(\text{select } i) = \frac{\exp(S_i^{(t)} / T_t)}{\sum_j \exp(S_j^{(t)} / T_t)}$$

Temperature annealing: T_t from **3.0** $\rightarrow$ **0.1**

Why Annealing Works

- **Early rounds** ($T = 3.0$): Near-uniform $\rightarrow$ explore all clients
- **Late rounds** ($T = 0.1$): Sharply picks best clients
- **Warmup** (3 rounds): Pure random to build estimates

Convolutional Prototype Network (CPN)

Input Pipeline

- 1 **Feature Selection:** Mutual information
→ top 15 features
- 2 **Standardization:** Zero mean, unit variance
- 3 **Square Projection:** $15D \rightarrow 5 \times 5$ grid (zero-pad to 25D)

Why CNN for Tabular Data?

- Can capture local metric correlations
- Very small (≈ 5000 parameters) — fast, resource-efficient
- Amplifies differences between selection strategies

CPN Layer Summary

Layer	Input	Output
Conv2D (8 filters, 3×3)	$1 \times 5 \times 5$	$8 \times 5 \times 5$
BatchNorm + ReLU + MaxPool	$8 \times 5 \times 5$	$8 \times 5 \times 5$
Conv2D (16 filters, 3×3)	$8 \times 2 \times 2$	$16 \times 2 \times 2$
BatchNorm + ReLU + MaxPool	$16 \times 2 \times 2$	$16 \times 2 \times 2$
Flatten + Linear $16 \rightarrow 32$	16	32
ReLU + Dropout (0.2)	32	32
Linear $32 \rightarrow C$	32	C

The 32D penultimate layer is the **embedding** used for prototypes.

Experimental Setup

9 Real-World Defect Datasets

Dataset	Samples	Defect Rate
Apache	194	15.5%
Safe	56	16.1%
Zxing	399	13.8%
CM1	327	9.8%
MW1	253	7.5%
PC1	705	6.9%
PC3	1077	10.2%
PC4	1287	12.2%
AEEEM	677	14.5%

Each dataset = one federated client. No raw data sharing.

Experimental Protocol

Rounds R	20
Local epochs E	5
Independent runs	10
Subset sizes k	$\{3, 5, 7, 8\}$
Prototype λ	0.2
Temp. $T_{\text{init}} \rightarrow T_{\text{final}}$	$3.0 \rightarrow 0.1$
Warmup W	3 rounds
UCB β	0.3

Evaluation Metrics

AUC · G-Mean · MCC · F1 · BalAcc

Statistical test: **paired t-test** (10 runs)

Significance: $p < 0.05$ (strong), $p < 0.10$ (marginal)

Main Results: Overall Performance at $k = 8$

Performance Comparison (mean $\pm$ std, 10 runs)

Algorithm	AUC	G-Mean	MCC	F1
Random	0.9856	0.6378	0.6269	0.9812
FedCS	0.9823	0.6214	0.6102	0.9789
PowerChoice	0.9871	0.6356	0.6301	0.9834
MultiMetric	0.9982	0.6423	0.6351	0.9921

Statistical Significance vs. Random ($k = 8$)

Metric	Improve	p-value	Sig.?
AUC	+1.28%	0.037	Yes
F1	+1.11%	0.025	Yes
MCC	+1.31%	0.087	Marginal
G-Mean	+0.71%	0.091	Marginal

Key Takeaways

- MultiMetric is **best on all metrics**
- FedCS is **worst** — volume bias hurts small but valuable clients
- Gains are **statistically significant** (not luck)
- Small absolute gains are **meaningful at this AUC level** (near ceiling)

Convergence Speed (AUC at Round 10)

Random	0.9512
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Ablation Study & Subset Size Analysis

Ablation: What does each signal contribute?

AUC when removing one component ($k = 8$)

Configuration	AUC
Full MultiMetric	0.9982
w/o Delta (70% signal removed)	0.9876
w/o Diversity	0.9901
w/o UCB	0.9956
w/o Consistency	0.9968
Only Delta (no others)	0.9892

Delta is most critical (−1.06% without it). **All four signals help** — using only delta performs

Performance across subset sizes

MultiMetric AUC by k (mean $\pm$ std)

k	AUC	MCC
3	0.9978 $\pm$ 0.0032	0.6312
5	0.9980 $\pm$ 0.0030	0.6338
7	0.9981 $\pm$ 0.0028	0.6345
8	0.9982 $\pm$ 0.0029	0.6351

Insight: Small k = biggest gain

At $k = 3$, only 3 of 9 clients are selected each round. **Intelligent selection matters most** when participation is limited.

Conclusions

What We Built

- **FPLPA**: Federated Prototype Learning with Privacy Awareness
- **MultiMetric**: Novel 4-signal client selection algorithm
- **CPN**: Lightweight CNN for resource-constrained clients
- Evaluated on **9 real-world datasets** with full statistical rigor

Key Findings

- MultiMetric **outperforms all baselines** across all k and metrics
- Improvements are **statistically significant**

Future Work

- **Differential privacy**: Add Gaussian noise to prototypes for formal (ϵ, δ) -DP guarantees
- **Adaptive weights**: Meta-learn w_{Δ} , w_{cons} , w_{div} online
- **Larger models**: Test with transformers or GNNs for defect prediction
- **Asynchronous FL**: Handle straggler clients
- **More datasets**: Bugzilla, GitHub Issues, industrial codebases

Key Literature

#	Authors (Year)	Paper	Relevance to Our Work
1	McMahan et al. (2017)	Communication-Efficient FL (FedAvg)	Foundation for our federated training loop
2	Nishio & Yonetani (2019)	FedCS: Client Selection with Heterogeneous Resources	Volume-based baseline we improve upon
3	Cho et al. (2020)	Biased Client Selection in FL (PowerChoice)	Loss-based baseline; we show multi-signal is better
4	Tan et al. (2022)	FedProto: Federated Prototype Learning	Prototype aggregation mechanism; $\lambda = 0.2$ from here
5	Zhang et al. (2021)	DivFL: Diverse Client Selection	Diversity signal inspiration for MultiMetric
6	Chicco & Jurman (2020)	MCC over F1 for Binary Classification	Motivated our use of MCC as primary imbalance metric

Speaker note: Use this slide if asked about related work during Q&A. You don't need to present all of it — just reference as needed.

Thank You!

Questions & Discussion

Project Title: FPLPA with MultiMetric Client Selection

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Best result: AUC **0.9982**, MCC **0.6351** ($k = 8$, $p = 0.037$)
